

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: Bachelor of Technology

Branch : Biomedical Engineering

Semester : VI

Subject Code & Name: BTBM601 & Medical Image Processing and Analysis

Max Marks: 60

Date: 18/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

Q. 1	Objective type questions. (Compulsory Question)				(Level/CO)	Marks
1	The first step in Image Processing is:					1
	a. Segmentation	b. Image acquisition	c. Image enhancement	d. Image restoration	L1/CO1	
2	The total pixels in any given region defines the:				L1/CO1	1
	a. Brightness	b. Intensity	c. Perimeter	d. Area		
3	The principal energy source for images in use today is the ____.				L1/CO1	1
	a. x-ray	b. B-ray	c. electromagnetic energy spectrum	d. laser ray		
4	Intensity transformations operates on _____				L1/CO2	1
	a. single pixel	b. two pixels	c. multiple pixels	d. all together in an image		
5	Spatial domain processes is expressed as				L1/CO2	1
	a. $g(x,y) = T[g(x,y)]$	b. $g(x,y) = T[f(x,y)]$	c. $f(x,y) = T[g(x,y)]$	d. none		
6	The negative of an image with intensity levels in the range from				L1/CO2	1
	a. [1, L-1]	b. [2, L-1]	c. [3, L-1]	d. [0, L-1]		
7	Which one is the abbreviation of LoG?				L1/CO3	1
	a. Laplacian of Gaussian	b. Laplacian of Gauss	c. Laplacian of Gauge	d. None		
8	The length of the run's is 0 are coded is called _____				L1/CO3	1
	a. Run length	b. Huffman coding	c. Compression	d. Modulation		
9	Edge detection is primarily used to				L1/CO1	1
	a. Detect boundaries between objects in an image	b. Measure the color intensity of an image	c. Smooth out the image	d. Enhance the brightness of an image		
10	Thresholding is a technique in image processing used to:				L1/CO4	1
	a. Smooth an image by reducing high-frequency noise	b. Segment an image into different regions based on pixel intensity	c. Enhance the color contrast of an image	d. Perform edge detection		

11	The Least Mean Square filter is particularly effective in:				L1/CO5	1
	a. Image segmentation	b. Removing noise from images	c. Enhancing the edges of an image	d. Detecting patterns in images		
12	Common causes of image degradation include				L1/CO5	1
	a. Over-exposure of the image	b. Noise, blur, and distortions during acquisition or transmission	c. Using color-based segmentation	d. Over-enhancement of image features		
Q. 2	Solve the following.					12
A)	With appropriate equation explain the simple image formation model.				L1/CO1	6
B)	Write a short note on : (i) Sampling (ii) Qunatization				L1/CO1	6
Q.3	Solve the following.					12
A)	With relevant explain basic transformation of gray scale image				L1/CO1	6
B)	Briefly explain the histogram processing works.				L1/CO1	6
Q. 4	Solve Any Two of the following.					12
A)	Write down the algorithm for canny edge detection for an image.				L2/CO1	6
B)	Using entire image region explain the basic formula for segmentation.				L1/CO1	6
C)	Briefly explain basic global thresholding in an image.				L1/CO1	6
Q.5	Solve Any Two of the following.					12
A)	Briefly explain about Huffman encoding for image compression				L1/CO1	6
B)	With the neat block diagram explain lossy predictive coding.				L2/CO1	6
C)	With an example briefly explain run length code?				L2/CO1	6
Q. 6	Solve Any Two of the following.					12
A)	Explain the model of the image degradation process.				L1/CO1	6
B)	How input image recovered by inverse filter.				L1/CO1	6
C)	Illustrate with an equation about Radon transform.				L4/CO1	6
*** End ***						